

● TECHNICAL DEMONSTRATION • REPRODUCIBLE PROTOTYPE

AI-BASED INERTIAL NAVIGATION

for UAVs in GPS-denied airspace

When GPS is **jammed, denied, or spoofed**, the aircraft must navigate on sensors an adversary cannot corrupt. A lightweight neural network learns the IMU's systematic error and removes it before a classical filter runs — **keeping heading and position bounded with no external signal and no magnetometer.**

-98%

ATTITUDE DRIFT

3×

TIGHTER @ 60 S DENIAL

~19k

PARAMS • EDGE-READY

0

MAGNETOMETER • JAM-PROOF



01 • THE THREAT

GPS is the single point of failure in autonomous flight.

In contested airspace, satellite navigation is routinely attacked. When the fix disappears, the UAV falls back on its Inertial Measurement Unit — and raw inertial navigation **diverges within seconds**.

MODE 01

Jammed

The signal is drowned out by hostile emitters. Routine in modern EW environments and contested logistics corridors.

MODE 02

Denied

The signal is absent — indoor, urban canyon, terrain-masked, or deliberately withheld theatre-wide.

MODE 03

Spoofed

The signal is falsified. The aircraft is confidently steered off course by an adversary. Trust in GNSS is gone.

RAW IMU DEAD-RECKONING • 2.5 MIN FLIGHT

33 km

POSITION ERROR (MAX 74 KM)

RAW ATTITUDE DRIFT • SAME FLIGHT

114°

ORIENTATION RMSE

ERROR GROWTH FROM GYRO BIAS

t^3

CUBIC POSITION DIVERGENCE

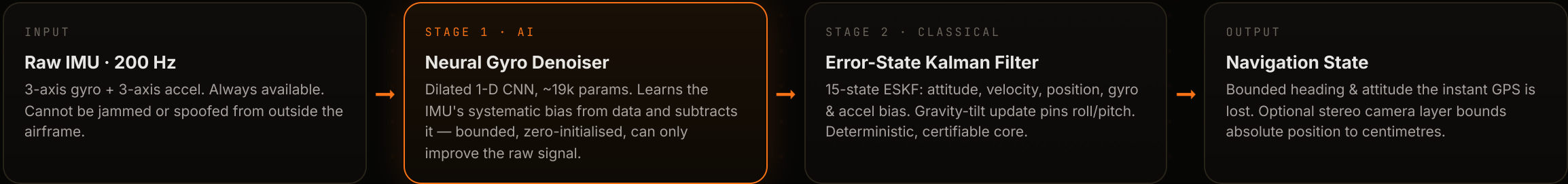
CORE
PHYSICS

Heading (yaw) is **physically unobservable** to an IMU-only filter — gravity pins roll & pitch but carries no heading information. Any gyro yaw-bias integrates into **unbounded heading error**, and a wrong heading steers the entire trajectory off course.

02 • THE SOLUTION

Learn the sensor's error. Then filter.

A two-stage architecture: a **~19k-parameter neural denoiser** removes the systematic gyro error the filter is structurally blind to — including the unobservable yaw bias — then a proven **15-state Error-State Kalman Filter** fuses the cleaned signal. Each stage validated in isolation.



$$\hat{\omega} = \omega_{\text{raw}} + \alpha \cdot \tanh(f_{\theta}(\text{window}))$$
 bounded correction, $\alpha = 0.15 \text{ rad/s}$ · zero-initialised: training starts at the raw gyro and can only improve it

DESIGN CHOICE • JAM-PROOF HEADING

No magnetometer. By design.

An adversary who can deny GPS can also spoof a magnetometer. Heading is held by the denoised gyro alone — verified in code: no heading measurement of any kind enters the filter.

OPTIONAL LAYER • APPROACH A

Camera bounds absolute position

OpenVINS stereo visual-inertial odometry rides on the same estimator core: **~6.2 cm** position error — a ~5000× reduction over inertial-only navigation.

03 • VALIDATED PERFORMANCE

Near-oracle accuracy. From data alone.

Strict cross-sequence hold-out on ground-truthed flight data (EuRoC MAV). The AI is never told the true bias — both sides start blind, the realistic GPS-denied case. Every number is the **mean over five independently-seeded retrains**.



ORIENTATION RMSE ON HELD-OUT FLIGHTS • DEGREES • 5 SEEDS

SEQUENCE	RAW ESKF	AI-NAV	ORACLE	GAIN
V1_02 • medium	76.0°	1.57°	0.75°	-97.9%
MH_02 • easy	90.2°	11.58°	5.08°	-87.2%
V2_03 • difficult	102.7°	3.44°	4.67°	-96.6%

On the hardest sequence the learned system **beats the oracle on every seed** — without ever being told the bias. The learned z-axis correction (−4.4°/s) matches the sensor's true yaw bias (+4.34°/s).

OPEN-LOOP ATTITUDE DRIFT

-98%

109.6° → 2.05° RMSE, HELD-OUT

FUSED POSITION ERROR

631 → 114 m

ATE, RAW VS AI • V1_02

TRAINING DETERMINISM

±1%

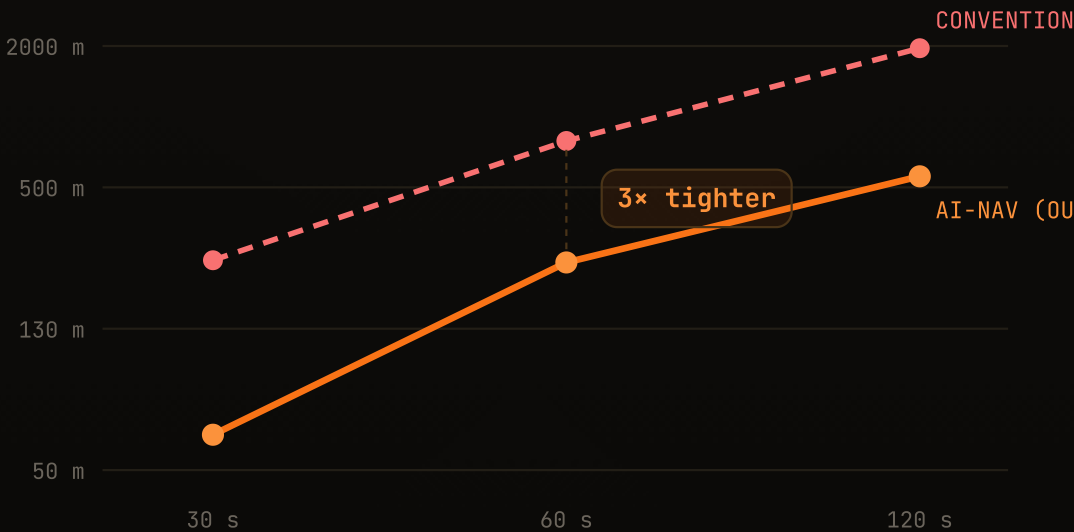
VAL. ERROR SPREAD ACROSS 5 SEEDS

04 • OPERATIONAL PAYOFF

GPS dies. How far off are you?

The question a defence buyer actually asks — answered as a **curve, not a point**. 20 GPS-cut instants swept across real flight; from each cut the system dead-reckons on inertial data alone. Median over all cuts. Identical filter both sides — the only difference is the AI cleaning the gyro first.

POSITION ERROR VS GPS-DENIAL TIME • MEDIAN, LOG SCALE



MEDIAN POSITION ERROR AFTER GPS CUT • V1_01, 20 CUT POINTS

DENIAL TIME	CONVENTIONAL INS	AI-NAV (OURS)	GAIN
30 s	290 m	67 m	+77%
60 s	781 m	288 m	+63%
120 s	1 704 m	582 m	+66%

HEADLINE • WELL-SAMPLED 60 S HORIZON

3x tighter

THAN CONVENTIONAL INERTIAL NAVIGATION AFTER ONE MINUTE OF GPS DENIAL — NO MAGNETOMETER, NO EXTERNAL SIGNAL

05 • FLIGHT DEMONSTRATOR

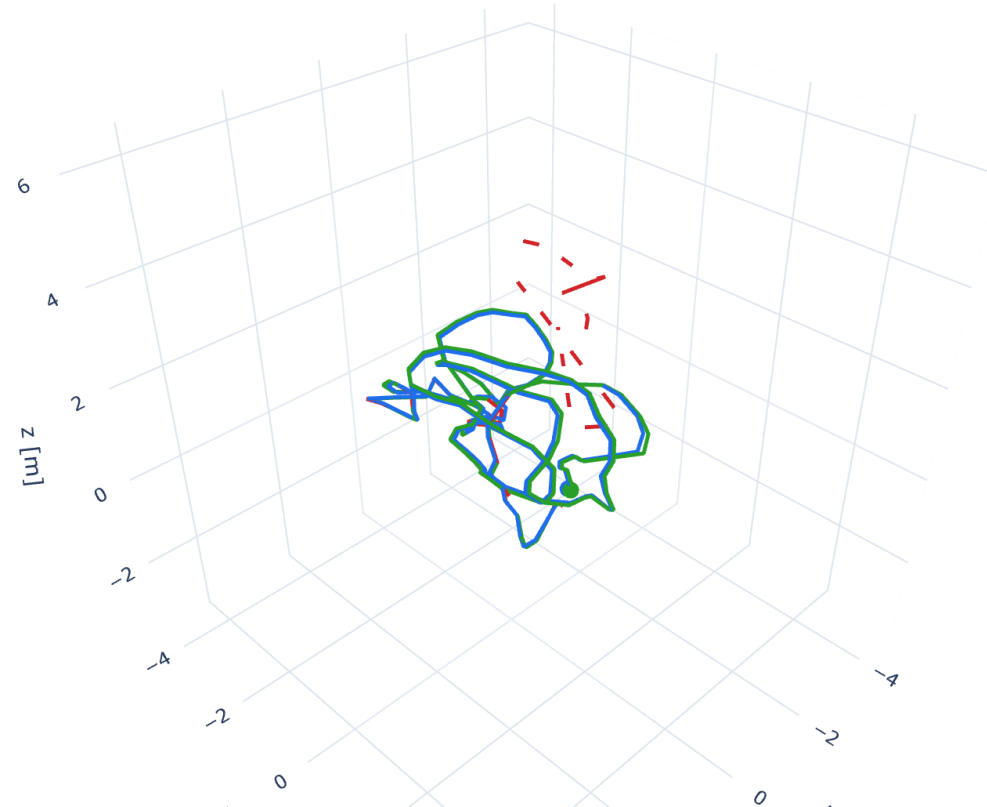
Live replay: the moment GPS is cut.

Real EuRoC flight, three estimators on one clock — final frame shown, $t = 143.6$ s. The conventional INS has ballooned to **309 m off course**; AI-Nav sits **6.3 cm from truth**. Interactive version in the HTML deck.

— GROUND
TRUTH— AI-NAV
(OURS)— INS • GPS-
DENIED

— Ground truth
— AI-Nav (ours)
— INS • GPS-denied

⚠ GPS DENIED

AI-NAV ERROR
6.3 cmINS DRIFT
309.2 mELAPSED
143.6 s

— 06 • HARDENED & HONEST

Stress-tested. Limits stated. Limits closed.

Three questions decide whether a result is trustworthy: does it survive a different random seed, a different environment, a different sensor? **All three tested directly** — and the failure modes are reported at full strength.

TEST 01 • MULTI-SEED

STABLE

5 independent retrains

Training is essentially deterministic: validation error $0.096^\circ \pm 0.001^\circ$ (~1% spread). The network reliably learns the same correction — the win is a property of the method, not a lucky checkpoint.

TEST 02 • CROSS-ENVIRONMENT

TRANSFERS

Train hall → test rooms

A network that never saw a Vicon room reaches 3.20° on the hardest room — beating the oracle. Reverse direction degrades gracefully, never fails: worst case still a 5× win over raw.

TEST 03 • CROSS-SENSOR

FINE-TUNE REQUIRED

The honest limit — and its fix

Frozen, the network does not transfer to a different IMU (it encodes one sensor's bias). A short per-IMU fine-tune on a physically different Bosch MEMS unit restores the win on never-seen data: $1.62^\circ \rightarrow 0.81^\circ$, beating raw (0.86°).

CAMERA LAYER • APPROACH A

6.2 cm

STEREO-VIO ABSOLUTE POSITION • ~5000× VS INERTIAL-ONLY

VALIDATION FOOTING

EuRoC • TUM-VI

PUBLIC, MM-ACCURATE GROUND TRUTH • FULLY REPRODUCIBLE REPO

SCIENTIFIC INTEGRITY

Negatives reported

NAIVE LEARNED VIO FRONT-END SCOPED & PUBLISHED, NOT HIDDEN

— 07 • PATH TO FIELDING

Three of four deployment gates closed.

The algorithm is proven and hardened on ground-truthed flight data. One step remains between this prototype and a fielded capability — and it needs hardware, not more analysis.

D4

Jam-proof heading — no magnetometer

Heading held by the denoised gyro alone; verified in code. A deliberate design choice: an adversary who denies GPS can also spoof a compass.

ADDRESSED

D2

New-sensor transfer

Per-IMU warm-start fine-tune demonstrated across a genuine hardware change (tactical-grade → consumer MEMS). Next: automate calibration from a short logged flight.

ADDRESSED

D3

Drift under sustained denial

Characterized as a quotable curve over 20 swept GPS-cut points: 3× tighter at 60 s, ~66% tighter at every horizon tested.

ADDRESSED

D1

Edge / realtime onboard deployment

~19k parameters + ESKF onto flight-controller-class compute (NVIDIA Jetson), quantized INT8, within per-sample latency budget. Parameter count and vectorized inference (44× speed-up already demonstrated in training) make the budget plausible.

NEXT • HARDWARE-GATED

The ask: target hardware & flight data.

Jetson-class embedded module + drone-grade IMU logs close D1 and carry a proven, honest, reproducible algorithm onto a flying platform.

INITIATE D1 →